ASM satellite detection in Côte d'Ivoire: build reference

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Context

The practical route is Sentinel-2 plus Sentinel-1 with a U-Net-style segmenter, trained on labels you build for Côte d'Ivoire, and the demand is real: officials and opposition politicians are still calling for a reliable site map, and two mapping projects started in 2026. Everything below is drawn from sources I retrieved on 22 Oct 2026 (search snippets, not full-text reads); items from my own general knowledge are marked.

- **Scale of the problem.** The environment ministry said in February 2026 that 29 of 31 regions are affected. The National Security Council reported on 23 July 2026 that more than 8,500 illegal sites were cleared between 2021 and mid-2026, and that activity is still spreading.
- **Government mapping is underway.** A mapping and land-rehabilitation project was launched at Daoukro on 27 Feb 2026 with Abidjan Legacy Program. A second effort started in May 2026 at Bouaflé with the NGO AEIE. Opposition politicians are also calling for an objective national map.
- **Existing Ivorian remote-sensing work.** The Bandama River study is the closest precedent: Sentinel-2 with AI models trained on field-campaign data, applied to about 600 km² of central Côte d'Ivoire (International Journal of Applied Earth Observation and Geoinformation, 2022). A separate INP-HB study at Kouassi-Périta measured roughly 70 ha of ASM in Nov 2017, 184+ new ha by Jan 2020 and 124+ more by Apr 2021.
- **Legal layer exists.** The e-cadastre minier portal launched on 22 Sep 2025. The ministry reports 213 valid artisanal mining authorisations, up from 157 in 2023. That is the licensed-versus-illegal reference you will want.

Best combination, and can it be daily?

The best stack is Sentinel-1 for change alerts, Sentinel-2 for confirmation and mapping, and a deep-learning segmenter on top, with PlanetScope only over priority areas. Daily processing is possible, but free satellites do not see any one place daily: expect a new

free observation about every 2 to 3 days on average, and a confirmed detection roughly weekly. Truly daily imagery needs paid PlanetScope.

Sensor	Revisit over Côte d'Ivoire	Role in the stack	Caveat
Sentinel-1 (C and D, radar)	6-day nominal revisit. Sentinel-1A was retired on 29 June 2026, leaving the C and D pair	Cloud-proof change alerts: new clearing, new ponds	Rain and terrain add noise; alerts are coarser than optical maps
Sentinel-2 (optical)	5 days at the equator from the two-satellite constellation	Confirmation, site boundaries, area statistics	Cloud: a CESBIO analysis says that, with about 70% average land cloud cover, Sentinel-2 often fails to give one clear view a month
PlanetScope (commercial)	Daily at nadir, 3 to 5 m	Daily watch and verification over hotspots	Paid; free NICFI access ended in January 2025. NASA's research programme releases it only 30 days after acquisition, so it cannot serve alerts

Fresh-observation cadence is my estimate from the revisit figures above: the two free sensors together give a new pass every 2 to 3 days, but optical passes are often cloud-blocked. Peru's RAMI alert system followed the same pattern, using Sentinel-1 alerts verified on PlanetScope daily imagery.

Service tiers to offer.

- **Daily run, free data.** A scheduled job checks for new scenes each day and processes whatever has arrived. Users see new SAR alerts about every 6 days per area and Sentinel-2 detections about every 5 days when clear.

- **Daily watch, paid data.** PlanetScope over a small set of hotspots, such as classified forests and river corridors flagged by alerts. Request a quote for the hotspot area before committing.
- **On-demand verification.** Order very-high-resolution images for flagged sites, as the ASM Alert platform does.

Data sources

Start with free Sentinel-2 and Sentinel-1 through Google Earth Engine or Planetary Computer; add very-high-resolution imagery only to confirm detections. Rows marked (GK) come from my general knowledge and were not looked up here, so confirm access terms before relying on them.

Data type	Use for ASM	Resolution / cadence	Access and notes
Sentinel-2 L2A (optical)	Main detector input; bare ground, turbid ponds, vegetation loss. Used in Ghana, Senegal, Burkina Faso and Côte d'Ivoire	10 m (visible, NIR), 20 m (SWIR); about 5-day revisit (GK)	Planetary Computer STAC, Google Earth Engine, Copernicus Data Space (GK)
Sentinel-1 GRD (C-band SAR)	Cloud-proof change detection; works under persistent cloud. A +1.65 dB backscatter threshold found new galamsey areas in Ghana	10 m; 6 to 12-day revisit (GK)	Google Earth Engine, ASF, Copernicus Data Space (GK). Terrain correction and DEM choice matter
Landsat 8/9 and archive	Long history; the Pará study ran LandTrendr on Landsat 2000 to 2019 in Earth Engine	30 m; 16-day (GK)	Earth Engine, USGS. Too coarse alone for small pits

Data type	Use for ASM	Resolution / cadence	Access and notes
ALOS PALSAR (L-band SAR)	Decade-long baseline; the Ghana Sentinel-1 ARD work compared against 2007/08 L-band mosaics	about 25 m (GK)	JAXA mosaics (GK)
PlanetScope / Tropical Forest Observatory	High-resolution confirmation and labelling. Free NICFI access ended in Jan 2025; the replacement is non-commercial and paid	about 4.7 m, monthly mosaics (GK)	A May 2025 forum post quotes 1,440 GBP per year; check current pricing
Commercial VHR (Pléiades, etc.)	Verification of flagged sites. Pléiades plus Sentinel-2 was used in Burkina Faso	0.5 to 2 m	Tasked or archive purchase; the ASM Alert platform lets users order VHR for flagged areas
Embeddings (AlphaEarth, SSL4EO, Prithvi, Clay)	Few-label classification on pretrained features. AlphaEarth gives annual 64-dimensional embeddings at 10 m from 2017 in Earth Engine (secondary source)	10 m, annual	Earth Engine; models on Hugging Face
Elevation and rivers (Copernicus DEM, HydroSHEDS, OSM)	Slope, distance to river, SAR terrain correction. Alluvial sites follow streams and rivers	30 m (GK)	Earth Engine (GK)

Data type	Use for ASM	Resolution / cadence	Access and notes
Forest loss and protected areas (Global Forest Watch, WDPA)	Prioritise detections inside classified forests and reserves	30 m (GK)	GFW still hosts Landsat and Sentinel-2 imagery
Rainfall (CHIRPS)	Screen out SAR backscatter noise from heavy rain; rain smearing is a known issue	about 5 km daily (GK)	Earth Engine (GK)
Mining cadastre	Separate licensed from unlicensed disturbance	Polygons	e-cadastre minier, built by Spatial Dimension; public data access to be confirmed with the ministry

Labels and benchmark datasets

No public, Côte d'Ivoire-specific ASM label set turned up, so plan to build one: bootstrap from global mine polygons, then digitise local sites and add field points. Global polygons mix industrial and artisanal mining and cover only part of the artisanal footprint.

Dataset	Contents	Relevance to Côte d'Ivoire	Access
Maus et al. global mining polygons v2	44,929 polygons, 101,583 km², from Sentinel-2 imagery for 2019; includes large-scale and artisanal mining	Starting layer. Mapping was limited to a 10 km buffer around 34,820 known mine coordinates, so unlisted artisanal sites are likely missing	PANGAEA, DOI 10.1594/PANGAEA.942325

Dataset	Contents	Relevance to Côte d'Ivoire	Access
Tang and Werner global mining footprint and the 1985 to 2022 time series	74,726 polygons, 82,552 km², with active and reclaimed status by year	Mine-versus-reclaimed history; delineated from high-resolution imagery, so finer than Maus	ESSD 2026 data record (link in the paper)
Jasansky and Maus ML-ready dataset	Sentinel-2 tile references and masks for 1,210 sites (2048 × 2048 px), tagged artisanal or industrial, with train/validation/test splits	Ready for training a segmenter; images are fetched from Planetary Computer, not bundled. Code: SimonJasansky/mine-segmentation	IIASA PURE page
MineNetCD	More than 70,000 paired before/after high-resolution patches from 100 sites, with change masks	Change-detection pretraining; mostly large mines. Code: EricYu97/MineNetCD	Rodare DOI 10.14278/rodare.3251; cropped version on Hugging Face
Couttenier et al. West Africa ASM map	1.75 million km² across 13 countries; sites from 0.1 ha to about 2,000 ha	Likely covers Côte d'Ivoire, but I could not confirm the data is public. Ask the authors. Related project: ArtiMining (ANR)	Unverified
Girard et al. replication package	"Artisanal mining in Africa. Green for Gold?", 50.4 GB, CC BY 4.0	Contents not checked; may include site locations for Africa	Zenodo

Dataset	Contents	Relevance to Côte d'Ivoire	Access
Amazon Mining Watch detections	Yearly 2018 to 2024 and quarterly from Q2 2025 mine-scar detections, CC BY 4.0	Different biome; useful for transfer learning and method comparison, not as labels	source.coop
Brazil Sentinel-2 patch dataset	2,600 patches (1,300 without mining, 1,300 across 10 mining-proportion bins)	Tests detection when mining is a small share of a patch	Workshop data paper
ELDOR UAV benchmark	Drone orthomosaics over 2,500+ ha with pixel labels for pits, rafts, machinery and regrowth	Shows the structures that satellites miss; Amazon, not West Africa	Hugging Face
Local ground truth	Field campaigns (used in the Bandama study), cadastre polygons, and clearance records. The government reports 1,114 sites cleared from January to July 2026	The most valuable and hardest to get. Ask INP-HB, the environment ministry or Abidjan Legacy Program for coordinates; label in QGIS or Collect Earth Online	Partnership needed

Research papers

Read the Bandama, Senegal and Couttenier papers first: together they give a local precedent, a classical GEE workflow and a West Africa-scale deep-learning baseline. Metrics are as stated in the abstracts I retrieved; read each paper before comparing numbers across studies.

Paper	Region	Data and method	What to take from it
Recent expansion of artisanal gold mining along the Bandama River (IJAEOG, 2022)	Central Côte d'Ivoire, about 600 km²	Sentinel-2; AI model trained on field-campaign data, adapted from the Senegal method	Local precedent; shows Sentinel-2 can quantify expansion rates
Spatio-temporal evolution of ASM in Central Côte d'Ivoire (INP-HB, 2022)	Kouassi-Périta, Yamoussoukro-Ouest	Multi-date satellite imagery, site extent by date	Growth figures useful as a sanity check on your area estimates
Mapping ASGM in Senegal using Sentinel-2 (GeoHealth, 2020)	South-east Senegal	PCA to pick the best season, SEaTH to pick bands and indices, SVM classifier, all in Earth Engine	Best season is the end of the dry season, when vegetation is lowest. Good classical baseline
Artisanal exploitation of mineral resources: remote sensing observations (Surveys in Geophysics, 2022)	West Africa review	Review of remote sensing for ASM impacts, illegal-site detection and expansion monitoring	Best single overview; includes social and ethical aspects
Mapping ASM at large scale from space with deep learning (PLoS ONE, 2022)	13 West African countries, 1.75 million km²	U-Net-style segmentation of Sentinel-2, November to February imagery	70% precision at 42% recall; sites from 0.1 to about 2,000 ha. Shows the recall gap you must plan around
Sentinel-2 multispectral CNN for ASM in Ghana (Remote Sensing of Environment, 2020)	Southern Ghana, 6 million ha	CNN on Sentinel-2, four-year change analysis	Over 3,500 ha of illegal-mining forest loss inside Ghana's protected forests, 2,400 ha since 2015

Paper	Region	Data and method	What to take from it
Mapping ASM change with machine and deep learning (Environmental Challenges, 2021)	Birim Basin, Ghana	ANN, random forest, SVM and pixel CNN on segmented Sentinel-2 bands	An ANN reached 99.80% overall accuracy; the study also ranks individual Sentinel-2 bands
ASM detection in semi-desert areas by improved U-Net (IEEE GRSL, 2022)	Burkina Faso	U-Net and attention-based U-Net on Sentinel-2, small incomplete inventory, dry and rainy seasons compared	Works with limited labels; season-by-season behaviour is analysed
Mapping small-scale mining in Ghana with Sentinel-1 time series (Remote Sensing, 2020)	South-west Ghana	Annual min/mean/max backscatter features, threshold change detection	+1.65 dB threshold; 102, 60 and 33 km² of new mining area in three successive periods
SAR for small-scale mining in Ghana (Stellenbosch thesis, 2020)	Ofin River, Ghana	Sentinel-1 and KompSAT-5 with object-based classification and differencing	SAR detects mining without commercial imagery; object-based analysis picked out mining water bodies
Change point detection for early-stage ASM (IJAEOG, 2025) and the 2026 PhD thesis	Wet southern Ghana	Smoothed Sentinel-1 series with CuSum, PELT and BinSeg change-point methods, then deep learning	Most relevant for early-warning alerts in rainy, cloudy conditions

Paper	Region	Data and method	What to take from it
Near-real-time alerts of illegal gold mining in Peru using SAR (Environmental Research Communications, 2024)	Peruvian Amazon	Sentinel-1 in Earth Engine with the Omnibus Q-test, validated on Planet imagery	Blueprint for an operational alert pipeline
Enhanced ASM detection with Landsat time series	Pará, Brazil	LandTrendr plus texture features (variance, contrast and others) in Earth Engine	Texture helps with mixed pixels at 30 m
MineNetCD (IEEE TGRS, 2024) and Prithvi-EO-2.0 (2024)	Global	Mining change-detection benchmark; geospatial foundation model for fine-tuning	Candidate pretrained backbones if labels are scarce

Projects, platforms and code

Three operational systems (Amazon Mining Watch, Peru's RAMI, ASM Alert) show the end product; the repositories give you code to start from.

Project	What it does	How to reuse it
Amazon Mining Watch (Earth Genome and partners)	Yearly and quarterly Sentinel-2 detections of artisanal gold-mine scars across the Amazon. From the 2024 data year it uses an ensemble of classification heads on SSL4EO ViT-DINO embeddings with human review. Code: the mining-detector repository	Closest open template for a patch-classifier on a foundation model. Note its warning: the model change makes 2023 to 2024 trends unreliable, so freeze your model before you report trends
RAMI, Peru (SERVIR-Amazonia, ACCA, Spatial Informatics Group)	Near-real-time Sentinel-1 alerts in Earth Engine, with a public web platform for enforcement agencies; early alerts validated in Collect Earth Online	Template for an alert workflow and a validation loop with human interpreters
ASM Alert (GAF and IES, funded by ESA)	Web platform combining Copernicus data, other geodata and machine learning to flag likely unlicensed ASM in tropical forests, with ordering of VHR imagery for flagged areas	Check whether the platform already covers Côte d'Ivoire before building a competing tool; possible partner
ASMSpotter (dida and RWTH Aachen)	Object detection of ASM sites on PlanetScope, labelled by a mining expert; planned as a cloud service	Shows the high-resolution route and its labelling cost
Ghana SERVIR and CERSGIS collaboration	User-driven mining-detection services in Peru and Ghana, with a planned technical exchange	Nearest West African operational context

Project	What it does	How to reuse it
Ghana Sentinel-1 ARD data cube	Sentinel-1 analysis-ready data over Ghana since March 2015, with galamsey monitoring as the lead use case	Model for a national analysis-ready SAR archive
SimonJasansky/mine-segmentation	Code that built the ML-ready mining dataset and trains segmentation models	Training scaffold on global mines
EricYu97/MineNetCD	Unified framework with more than a dozen change-detection models	Compare change-detection architectures
NASA-IMPACT/Prithvi-EO-2.0	Open geospatial foundation model weights and fine-tuning code	Fine-tune for mine segmentation when labels are few
GhanaArtisanalMining (Zenodo, 2021)	Small code archive (12 kB) for a Ghana artisanal-mining analysis	Worth reading; I did not check which paper it supports
IPIS open data	Field-visit datasets on artisanal mining sites in eastern DRC	Example of what a shared ground-truth dataset can look like

Implementation plan

Build a Sentinel-2 detector first, then add Sentinel-1 for alerts, and prove it on one pilot area before scaling. This sequence is my synthesis of the papers above, not a single published recipe.

1. **Scope the question.** Decide whether the output is a site inventory, yearly expansion statistics or near-real-time alerts. Each needs different data: Sentinel-2 composites for the first two, Sentinel-1 time series for the third.
2. **Pick a pilot area.** Use the Bandama corridor or the Daoukro and Bouaflé zones where government projects are running, so you can compare and partner. About 35% of West Africa's greenstone belts lie in Côte d'Ivoire, per the ministry's sector overview.

3. **Build the image stack.** Create cloud-masked Sentinel-2 L2A composites per season and year. Test a dry-season window first: the Senegal study found the end of the dry season best, and the West Africa study used November to February. Compute your own season choice for Côte d'Ivoire's wetter south. Add Sentinel-1 backscatter, DEM slope and distance to river as extra channels.
4. **Make labels.** Seed from Maus and the ML-ready dataset, then digitise local sites on Sentinel-2 and high-resolution imagery. Add field points if a partner can collect them. Label hard negatives on purpose: bare soil, quarries, roads, burn scars, farm clearings and naturally turbid rivers.
5. **Run a classical baseline.** Random forest or SVM in Earth Engine on spectral indices, as in the Senegal workflow. It gives a benchmark and a quick first map.
6. **Train a deep model.** U-Net-type segmentation on Sentinel-2 patches (Nava, Couttenier). If labels stay scarce, compare fine-tuning Prithvi-EO-2.0 or classifying patches on frozen embeddings, as Amazon Mining Watch does. Expect recall to be the weak point: the large-scale West Africa model reached 42%.
7. **Add SAR change alerts.** Smooth Sentinel-1 series and apply change-point or Omnibus tests; screen for rainfall noise. Use terrain-corrected, analysis-ready backscatter.
8. **Validate properly.** Hold out whole geographic blocks, not random pixels. Report precision, recall and per-site detection by size class. Estimate area from a stratified sample of independently interpreted points rather than from map pixel counts (general practice, not from these papers).
9. **Overlay the legal layer.** Intersect detections with the e-cadastre authorisations and classified forests to label each as licensed, unlicensed or in a protected area.
10. **Deliver.** Publish a web map and a vector layer. Route SAR alerts to human review before anything goes to the field, and freeze the model version for each reporting year so that trends stay comparable.

Development and implementation plan

Deliver a validated pilot map and alert feed in about seven months, then scale nationally over the following six. Durations, team size and targets below are my planning estimates, not figures from the sources; adjust them to your funding and staff.

Phase	Timing (estimate)	Main work	Deliverables	Exit gate
0. Mobilise	Weeks 0 to 4	Name the end users (environment ministry, mines directorate, Abidjan Legacy Program, INP-HB). Agree data sharing, access rules and who reviews alerts. Choose the pilot area	Signed partner roles, requirements note, pilot polygon, data-governance rules	Partners commit labels or field access
1. Foundation	Weeks 3 to 10	Set up Earth Engine or Planetary Computer access and a cloud workspace. Build Sentinel-2 composites and Sentinel-1 analysis-ready series. Digitise first labels, including hard negatives	Reproducible data pipeline, label set v1, label guide	Labels cover the pilot's main land-cover confusions
2. Models	Weeks 8 to 18	Random-forest baseline, U-Net segmenter, optional foundation-model fine-tune, SAR change-point or Omnibus alerts. Block cross-validation	Benchmark report with precision, recall and per-size results; model cards	Beat the baseline on held-out blocks; agree a target precision with users

Phase	Timing (estimate)	Main work	Deliverables	Exit gate
3. Pilot operations	Weeks 16 to 30	Daily scheduled run on the pilot, alert review screen, field or high-resolution verification of a sample of alerts, user feedback	Live alert feed, web map, verified-alert log, error analysis	Reviewed precision meets the agreed target for three consecutive months
4. Scale-up	Months 7 to 13	Extend to all gold belts, build the annual national inventory, add licensed versus unlicensed overlay from the cadastre, retrain with new labels	National site layer, yearly area statistics, retrained and frozen model v2	Independent accuracy check on a stratified sample
5. Operate	Month 13 onward	Monitoring, yearly retraining, budget and ownership handover, training for ministry staff	Runbook, handover, annual report	Ministry or partner runs it without the build team

Team (minimum). One remote-sensing lead, one machine-learning engineer, one data or cloud engineer, two or three image interpreters or field verifiers, and a part-time legal and ethics adviser.

Daily pipeline design.

1. A scheduler starts once a day and lists new Sentinel-1 and Sentinel-2 scenes over the area of interest.
2. New scenes are cloud-masked (optical) or terrain-corrected and speckle-filtered (radar).
3. Models run on the new data and on the latest composite; results are stored as cloud-optimised rasters and vector polygons.

4. Change logic compares the new result with the last known state and issues an alert only for new, persistent changes, not single noisy passes.
5. Alerts are intersected with the cadastre, classified forests and rainfall data, then sent to a review queue.
6. A human confirms or rejects each alert; the decision is saved as a new label for retraining.
7. Confirmed sites are published to the web map and the data service, with the model version recorded.

Proposed success measures. These are starting targets to agree with users: reviewed alert precision of at least 70%, median delay from satellite pass to published alert of 3 days or less, recall measured and reported by site size, and a model version frozen for each annual statistic.

Main cost drivers. Engineer time, labelling and field verification, cloud compute, and any commercial imagery. Get a PlanetScope quote for the hotspot area before choosing the paid daily tier; I did not find a current price I could verify.

Risks, gaps and what I could not verify

- **Small targets, mixed pixels.** At 10 m, pits and ponds under about 0.1 ha blend with surroundings. The West Africa model detected sites from 0.1 ha upward, yet its overall recall was only 42%.
- **Unstable trends.** A model change can look like growth in mining. Amazon Mining Watch says its 2024 model is more sensitive and that 2023 to 2024 trends are unreliable.
- **Fragile models.** A Brazilian study found that a single small adversarial perturbation turned every forest patch into a false mining detection. Test robustness before outputs feed enforcement.
- **Human consequences.** Detections may guide operations: the government reports 1,114 sites cleared from January to July 2026, and 65% of people referred to court since 2021 were foreign nationals. The review in Surveys in Geophysics lists about 503,100 people in Côte d'Ivoire in ASM in 2017 and discusses the social and ethical side. Agree data-sharing terms early, keep site-level layers access-controlled, and never present model output as proof of illegality.
- **Not verified.** I read search snippets, not full papers or data pages. I could not confirm that the Couttenier West Africa layer is public, what is in the Girard et al. Zenodo package, whether ASM Alert covers Côte d'Ivoire, or what the e-cadastre exposes as

open data. Rows marked (GK) in the data table are from general knowledge.

- **Cost.** Free NICFI Planet mosaics ended in January 2025; verify current Tropical Forest Observatory terms if you need monthly high-resolution imagery.